

5.3.7 External clock source characteristics

High-speed external user clock generated from an external source

In bypass mode the HSE oscillator is switched off and the input pin is a standard GPIO.

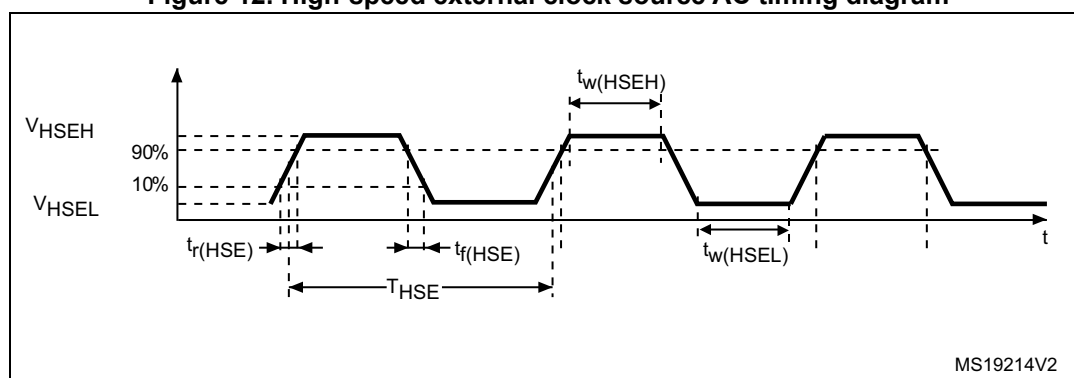
The external clock signal has to respect the I/O characteristics in [Section 5.3.13](#). See [Figure 12](#) for recommended clock input waveform.

Table 36. High-speed external user clock characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
$f_{\text{HSE_ext}}$	User external clock source frequency	-	-	8	48	MHz
V_{HSEH}	Digital OSC_IN input pin high level voltage	-	$0.7 \times V_{\text{DD}}$	-	V_{DD}	V
V_{HSEL}	Digital OSC_IN input pin low level voltage	-	V_{SS}	-	$0.3 \times V_{\text{DD}}$	
$t_{\text{w(HSEH)}}$ / $t_{\text{w(HSEL)}}$	Digital OSC_IN high or low time	-	7	-	-	ns

1. Specified by design. Not tested in production.

Figure 12. High-speed external clock source AC timing diagram



Low-speed external user clock generated from an external source

In bypass mode the LSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in [Section 5.3.13](#). See [Figure 13](#) for recommended clock input waveform.

Table 37. Low-speed external user clock characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
$f_{\text{LSE_ext}}$	User external clock source frequency	-	-	32.768	1000	kHz